

The Discrete Micropolar Problem: Inf-Sup Stability, Taylor–Hood Elements, and Error Estimates

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Structure. Part I states, without commentary, every definition and result established in Lessons 1 to 3. It is a reference sheet. Part II develops the new material, and each section there splits into a *Discussion* subsection carrying the explanation and motivation, and a *Mathematics* subsection carrying the statements and proofs.

Part I

Lessons 1 to 3, stated mathematically

1 Notation

$\Omega \subset \mathbb{R}^2$ bounded Lipschitz, boundary $\partial\Omega$, outward unit normal $\mathbf{n} = (n_1, n_2)$. All derivatives weak. $\partial_1 = \partial/\partial x$, $\partial_2 = \partial/\partial y$.

$$\begin{aligned}\nabla w &= (\partial_1 w, \partial_2 w), & \nabla^\perp w &= (\partial_2 w, -\partial_1 w), & \Delta &= \partial_1^2 + \partial_2^2, \\ \operatorname{div} \mathbf{v} &= \partial_1 v_1 + \partial_2 v_2, & \operatorname{curl} \mathbf{v} &= \partial_1 v_2 - \partial_2 v_1, \\ (\phi, \psi) &= \int_\Omega \phi \psi \, dx, & (\phi, \psi) &= \int_\Omega \phi \cdot \psi \, dx.\end{aligned}$$

2 Spaces

$$L^2(\Omega) = \left\{ \phi \text{ measurable} : \int_\Omega |\phi|^2 \, dx < \infty \right\} / \sim_{\text{a.e.}}, \quad \|\phi\|_{L^2} = \left(\int_\Omega |\phi|^2 \, dx \right)^{1/2}, \quad (1)$$

$$L_0^2(\Omega) = \left\{ q \in L^2(\Omega) : \int_\Omega q \, dx = 0 \right\}, \quad (2)$$

$$H^1(\Omega) = \left\{ \phi \in L^2(\Omega) : \nabla \phi \in [L^2(\Omega)]^2 \right\}, \quad \|\phi\|_{H^1}^2 = \|\phi\|_{L^2}^2 + \|\nabla \phi\|_{L^2}^2, \quad (3)$$

$$H_0^1(\Omega) = \overline{C_c^\infty(\Omega)}^{\|\cdot\|_{H^1}} = \left\{ \phi \in H^1(\Omega) : \phi|_{\partial\Omega} = 0 \right\}. \quad (4)$$

$$\mathbf{V} = [H_0^1(\Omega)]^2, \quad W = H_0^1(\Omega), \quad Q = L_0^2(\Omega), \quad \mathbf{V}_{\operatorname{div}} = \{ \mathbf{v} \in \mathbf{V} : \operatorname{div} \mathbf{v} = 0 \}. \quad (5)$$

$$\|(\mathbf{v}, z)\|^2 = \|\mathbf{v}\|_{H^1}^2 + \|z\|_{H^1}^2, \quad \|(\mathbf{v}, z, q)\|_X^2 = \|\mathbf{v}\|_{H^1}^2 + \|z\|_{H^1}^2 + \|q\|_{L^2}^2. \quad (6)$$

$$L^2(\Omega) = L_0^2(\Omega) \oplus \{\text{constants}\} \quad \text{orthogonally.} \quad (7)$$

3 Integration by parts

For ϕ, ψ scalar, $i \in \{1, 2\}$:

$$\int_{\Omega} (\partial_i \phi) \psi \, dx = \int_{\partial\Omega} \phi \psi n_i \, ds - \int_{\Omega} \phi (\partial_i \psi) \, dx. \quad (8)$$

For $\phi \in H^2(\Omega)$, $\psi \in H^1(\Omega)$:

$$\int_{\Omega} (\Delta \phi) \psi \, dx = \int_{\partial\Omega} \frac{\partial \phi}{\partial n} \psi \, ds - \int_{\Omega} \nabla \phi \cdot \nabla \psi \, dx. \quad (9)$$

For $\mathbf{v} \in [H^1(\Omega)]^2$:

$$\int_{\Omega} \operatorname{div} \mathbf{v} \, dx = \int_{\partial\Omega} \mathbf{v} \cdot \mathbf{n} \, ds. \quad (10)$$

4 Inequalities and abstract theorems

Lemma 4.1 (Cauchy–Schwarz). $|(\phi, \psi)| \leq \|\phi\|_{L^2} \|\psi\|_{L^2}$ for $\phi, \psi \in L^2(\Omega)$.

Lemma 4.2 (Poincaré–Friedrichs). $\exists C_{\Omega} > 0$ with $\|\phi\|_{L^2} \leq C_{\Omega} \|\nabla \phi\|_{L^2}$ for all $\phi \in H_0^1(\Omega)$; componentwise for $\mathbf{v} \in [H_0^1(\Omega)]^2$.

Lemma 4.3 (Young with parameter). $|ab| \leq \frac{\varepsilon}{2} a^2 + \frac{1}{2\varepsilon} b^2$ for all $a, b \in \mathbb{R}$ and all $\varepsilon > 0$.

Lemma 4.4 (Rotation identity). $(\nabla^\perp w, \mathbf{v}) = (w, \operatorname{curl} \mathbf{v})$ for $w \in H_0^1(\Omega)$, $\mathbf{v} \in [H_0^1(\Omega)]^2$.

Lemma 4.5 (Curl–divergence decomposition). For $\mathbf{v} \in [H_0^1(\Omega)]^2$: $\|\nabla \mathbf{v}\|_{L^2}^2 = \|\operatorname{curl} \mathbf{v}\|_{L^2}^2 + \|\operatorname{div} \mathbf{v}\|_{L^2}^2$, hence $\|\operatorname{curl} \mathbf{v}\|_{L^2} \leq \|\nabla \mathbf{v}\|_{L^2}$.

Lemma 4.6 (Divergence bound). $\|\operatorname{div} \mathbf{v}\|_{L^2} \leq \sqrt{2} \|\nabla \mathbf{v}\|_{L^2}$ for $\mathbf{v} \in [H^1(\Omega)]^2$.

Theorem 4.7 (Lax–Milgram). V Hilbert, a bilinear with $|a(u, v)| \leq M \|u\|_V \|v\|_V$ and $a(u, u) \geq \alpha \|u\|_V^2$, $F \in V'$. Then there is a unique $u \in V$ with $a(u, v) = F(v)$ for all $v \in V$, and $\|u\|_V \leq \alpha^{-1} \|F\|_{V'}$.

Theorem 4.8 (Brezzi). V, Q Hilbert, a bounded on $V \times V$, b bounded on $V \times Q$, $Z = \{v \in V : b(v, q) = 0 \, \forall q \in Q\}$. If

(B1) $a(v, v) \geq \alpha \|v\|_V^2$ for all $v \in Z$, and

(B2) $\inf_{q \in Q \setminus \{0\}} \sup_{v \in V \setminus \{0\}} \frac{b(v, q)}{\|v\|_V \|q\|_Q} \geq \beta > 0$,

then the saddle point problem has a unique solution depending continuously on the data.

5 The micropolar problem

5.1 Strong form

$$-(\nu + \nu_r) \Delta \mathbf{u} + \nabla p = 2\nu_r \nabla^\perp w + \mathbf{f}, \quad (11)$$

$$\operatorname{div} \mathbf{u} = 0, \quad (12)$$

$$-(c_a + c_d) \Delta w + 4\nu_r w = 2\nu_r \operatorname{curl} \mathbf{u} + g, \quad (13)$$

with $\mathbf{u} = \mathbf{0}$, $w = 0$ on $\partial\Omega$, and $\nu > 0$, $c_a + c_d > 0$, $\nu_r \geq 0$.

5.2 Bilinear forms

$$a((\mathbf{u}, w), (\mathbf{v}, z)) = (\nu + \nu_r) (\nabla \mathbf{u}, \nabla \mathbf{v}) + (c_a + c_d) (\nabla w, \nabla z) + 4\nu_r (w, z) - 2\nu_r (\nabla^\perp w, \mathbf{v}) - 2\nu_r (\operatorname{curl} \mathbf{u}, z), \quad (14)$$

$$b(\mathbf{v}, q) = - \int_{\Omega} q \operatorname{div} \mathbf{v} \, dx, \quad F(\mathbf{v}, z) = (\mathbf{f}, \mathbf{v}) + (g, z). \quad (15)$$

5.3 Mixed weak formulation

Find $((\mathbf{u}, w), p) \in (\mathbf{V} \times W) \times Q$ with

$$a((\mathbf{u}, w), (\mathbf{v}, z)) + b(\mathbf{v}, p) = F(\mathbf{v}, z) \quad \forall (\mathbf{v}, z) \in \mathbf{V} \times W, \quad (16)$$

$$b(\mathbf{u}, q) = 0 \quad \forall q \in Q. \quad (17)$$

5.4 Established results

$$|a((\mathbf{u}, w), (\mathbf{v}, z))| \leq M \|(\mathbf{u}, w)\| \|(\mathbf{v}, z)\|, \quad M = \nu + (c_a + c_d) + 9\nu_r. \quad (18)$$

$$a((\mathbf{v}, z), (\mathbf{v}, z)) \geq \alpha \|(\mathbf{v}, z)\|^2 \quad \forall (\mathbf{v}, z) \in \mathbf{V} \times W, \quad \alpha = \frac{\min(\nu, c_a + c_d)}{C_{\Omega}^2 + 1}, \quad (19)$$

obtained with $\varepsilon = 1/2$ in Lemma 4.3.

$$|b(\mathbf{v}, q)| \leq \sqrt{2} \|\mathbf{v}\|_{H^1} \|q\|_{L^2}. \quad (20)$$

$$Z := \{(\mathbf{v}, z) \in \mathbf{V} \times W : b(\mathbf{v}, q) = 0 \, \forall q \in Q\} = \mathbf{V}_{\operatorname{div}} \times W. \quad (21)$$

$$\mathcal{A}((\mathbf{u}, w, p), (\mathbf{v}, z, q)) := a((\mathbf{u}, w), (\mathbf{v}, z)) + b(\mathbf{v}, p) - b(\mathbf{u}, q) \quad \text{is not coercive on } X. \quad (22)$$

$$\inf_{q \in Q \setminus \{0\}} \sup_{\mathbf{v} \in \mathbf{V} \setminus \{0\}} \frac{b(\mathbf{v}, q)}{\|\mathbf{v}\|_{H^1} \|q\|_{L^2}} \geq \beta > 0 \quad (\text{Bogovskii, Nečas}). \quad (23)$$

$$b(\mathbf{v}, 1) = 0 \quad \forall \mathbf{v} \in \mathbf{V} \quad \implies \quad (23) \text{ fails if } Q \text{ is replaced by } L^2(\Omega). \quad (24)$$

By (18), (19), (20), (21), (23) and Theorem 4.8, problem (16)–(17) is well posed.

Part II

The discrete problem

6 Discretisation

6.1 Discussion

Part I establishes that the continuous mixed problem has a unique solution. Nothing there can be computed, because the spaces are infinite dimensional. To compute we replace $\mathbf{V}$, W , Q by finite dimensional subspaces, which turns the problem into a linear system in finitely many unknowns.

The finite element construction does this by dividing Ω into small triangles and taking functions that are polynomial on each triangle and continuous across edges. Two parameters control the approximation: the mesh size h , the diameter of the largest triangle, and the polynomial degree k . Refining the mesh means letting $h \rightarrow 0$.

There is a condition on how the triangles may degenerate as $h \rightarrow 0$, called shape regularity. Without it the triangles may become arbitrarily thin and the constants in the interpolation estimates of Section 10 blow up. The condition bounds the ratio of each triangle's diameter to that of its inscribed circle, uniformly over the whole family of meshes. It is a restriction on the mesh generator rather than on the analysis, and any reasonable mesh satisfies it.

Continuity across edges matters because the velocity and microrotation spaces must be subspaces of $H^1(\Omega)$. A piecewise polynomial with a jump across an edge is not in $H^1(\Omega)$: its weak derivative contains a Dirac measure supported on the edge, which is not an $L^2(\Omega)$ function. So continuity is not an aesthetic preference, it is exactly the condition for conformity. The pressure space is different, since $Q = L_0^2(\Omega)$ involves no derivatives and discontinuous pressures would be admissible. We nevertheless use continuous pressures, for reasons appearing in Section 9.

6.2 Mathematics

Definition 6.1 (Triangulation and shape regularity). $\mathcal{T}_h$ partitions $\bar{\Omega}$ into closed triangles meeting edge to edge. Write $h_K = \text{diam}(K)$, ρ_K for the diameter of the inscribed circle of K , and $h = \max_{K \in \mathcal{T}_h} h_K$. A family $\{\mathcal{T}_h\}_{h>0}$ is shape regular if

$$\exists \sigma > 0 : \quad \frac{h_K}{\rho_K} \leq \sigma \quad \forall K \in \mathcal{T}_h, \forall h > 0. \quad (25)$$

Definition 6.2 (Polynomial spaces).

$$\mathbb{P}_k(K) = \{\text{polynomials of total degree } \leq k \text{ on } K\}, \quad \dim \mathbb{P}_k(K) = \frac{1}{2}(k+1)(k+2),$$

$$\mathcal{L}_k = \{\phi \in C(\bar{\Omega}) : \phi|_K \in \mathbb{P}_k(K) \ \forall K \in \mathcal{T}_h\}, \quad \mathcal{L}_k^0 = \mathcal{L}_k \cap H_0^1(\Omega).$$

Proposition 6.3 (Conformity). $\mathcal{L}_k \subset H^1(\Omega)$ and $\mathcal{L}_k^0 \subset H_0^1(\Omega)$.

Proof. A piecewise polynomial has classical derivatives on each K , bounded there, hence in $L^2(\Omega)(K)$; summing over the finitely many K gives an $L^2(\Omega)(\Omega)$ function. Continuity across edges ensures this function is the weak derivative: in (8) the contributions from an interior edge shared by two triangles carry opposite normals and cancel exactly when the trace is single valued. Without continuity they do not cancel and no $L^2(\Omega)$ weak derivative exists. $\square$

Definition 6.4 (Taylor–Hood with $\mathbb{P}_2$ microrotation).

$$\mathbf{V}_h = [\mathcal{L}_2^0]^2 \subset \mathbf{V}, \quad W_h = \mathcal{L}_2^0 \subset W, \quad Q_h = \mathcal{L}_1 \cap L_0^2(\Omega) \subset Q. \quad (26)$$

Definition 6.5 (Discrete problem). Find $((\mathbf{u}_h, w_h), p_h) \in (\mathbf{V}_h \times W_h) \times Q_h$ with

$$a((\mathbf{u}_h, w_h), (\mathbf{v}_h, z_h)) + b(\mathbf{v}_h, p_h) = F(\mathbf{v}_h, z_h) \quad \forall (\mathbf{v}_h, z_h) \in \mathbf{V}_h \times W_h, \quad (27)$$

$$b(\mathbf{u}_h, q_h) = 0 \quad \forall q_h \in Q_h. \quad (28)$$

Definition 6.6 (Discrete kernel).

$$Z_h = \{(\mathbf{v}_h, z_h) \in \mathbf{V}_h \times W_h : b(\mathbf{v}_h, q_h) = 0 \ \forall q_h \in Q_h\}.$$

7 What transfers to the discrete spaces

7.1 Discussion

Having chosen $\mathbf{V}_h, W_h, Q_h$ as subspaces, the question is which of the results (18) to (23) survive when the spaces are replaced by their discrete counterparts. The answer is not uniform, and this is the central point of the lesson.

Upper bounds transfer with no argument at all. An inequality valid for every element of a set is valid for every element of a subset. So boundedness of a and of b hold discretely with identical constants.

Coercivity is more delicate. Brezzi's theorem requires coercivity on the kernel, and the discrete kernel is not contained in the continuous one. Membership of Z_h requires $\operatorname{div} \mathbf{v}_h$ to be orthogonal only to the finitely many functions in Q_h , whereas membership of Z requires orthogonality to all of Q , which forces the divergence to vanish. So discretely divergence free is strictly weaker than divergence free, and elements of Z_h generally have nonzero divergence. Coercivity proved only on Z would therefore say nothing about Z_h .

For the micropolar operator this difficulty does not arise, because the Lesson 2 coercivity proof never used the constraint. Every step of it, namely the identity from setting the arguments equal, Cauchy–Schwarz, Young with $\varepsilon = 1/2$, the curl bound, and Poincaré, is valid on all of $\mathbf{V} \times W$. This is exactly why Lemma 4.5 was recorded as an inequality rather than as the equality holding on $\mathbf{V}_{\operatorname{div}}$. So coercivity transfers and condition (B1) is free.

The inf-sup condition does not transfer, and this is the whole difficulty. The condition asserts that a supremum over $\mathbf{V}$ is bounded below. Discretely the supremum is over the smaller set $\mathbf{V}_h$, and restricting the set over which a supremum is taken can only decrease it. So the continuous condition yields an upper bound on the discrete quantity, which is useless, since a lower bound is what is needed. There is a competing effect, in that the infimum is also over the smaller set Q_h , which helps. Whether the two effects balance depends entirely on the relationship between $\mathbf{V}_h$ and Q_h , and this is why the element pair cannot be chosen freely.

7.2 Mathematics

Proposition 7.1 (Boundedness transfers). *Estimates (18) and (20) hold on $\mathbf{V}_h \times W_h$ and Q_h with the same constants M and $\sqrt{2}$.*

Proof. $\mathbf{V}_h \times W_h \subset \mathbf{V} \times W$ and $Q_h \subset Q$, and the estimates hold for all elements of the larger spaces. $\square$

Proposition 7.2 (The kernels differ). *In general $Z_h \not\subset Z$.*

Proof. By (21), $(\mathbf{v}, z) \in Z$ if and only if $\operatorname{div} \mathbf{v} = 0$ almost everywhere. Membership of Z_h requires only $\operatorname{div} \mathbf{v}_h \perp Q_h$ in $L^2(\Omega)$. For $\mathbf{v}_h \in [\mathcal{L}_2^0]^2$ the function $\operatorname{div} \mathbf{v}_h$ is piecewise linear but in general discontinuous across edges, whereas Q_h consists of continuous piecewise linears. Hence $\operatorname{div} \mathbf{v}_h$ generally has a nonzero component orthogonal to Q_h , and $\mathbf{v}_h \in Z_h$ does not force $\operatorname{div} \mathbf{v}_h = 0$. $\square$

Proposition 7.3 (Coercivity transfers).

$$a((\mathbf{v}_h, z_h), (\mathbf{v}_h, z_h)) \geq \alpha \|(\mathbf{v}_h, z_h)\|^2 \quad \forall (\mathbf{v}_h, z_h) \in \mathbf{V}_h \times W_h,$$

with α as in (19). In particular (B1) holds on Z_h .

Proof. Estimate (19) was proved on all of $\mathbf{V} \times W$, its proof using only Lemmas 4.1, 4.3, 4.4, 4.5 and 4.2, none of which requires $\operatorname{div} \mathbf{v} = 0$. Since $Z_h \subset \mathbf{V}_h \times W_h \subset \mathbf{V} \times W$, the estimate holds on both. $\square$

Proposition 7.4 (Inf-sup does not transfer). *Let*

$$S_h := \inf_{q_h \in Q_h \setminus \{0\}} \sup_{\mathbf{v}_h \in \mathbf{V}_h \setminus \{0\}} \frac{b(\mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1} \|q_h\|_{L^2}}.$$

Then for each fixed $q_h \in Q_h \setminus \{0\}$,

$$\sup_{\mathbf{v}_h \in \mathbf{V}_h} \frac{b(\mathbf{v}_h, q_h)}{\|\mathbf{v}_h\|_{H^1} \|q_h\|_{L^2}} \leq \sup_{\mathbf{v} \in \mathbf{V}} \frac{b(\mathbf{v}, q_h)}{\|\mathbf{v}\|_{H^1} \|q_h\|_{L^2}},$$

so (23) gives no lower bound on S_h .

Proof. $\mathbf{V}_h \subset \mathbf{V}$, and the supremum of a fixed quantity over a subset is at most the supremum over the whole set. Condition (23) bounds the right hand side below, which is compatible with the left hand side being arbitrarily small. $\square$

Definition 7.5 (Uniform discrete inf-sup). The pair $(\mathbf{V}_h, Q_h)$ is inf-sup stable if

$$\exists \beta_* > 0 \text{ independent of } h : \quad S_h \geq \beta_* \quad \forall h > 0. \quad (29)$$

8 Failure of the equal order pair

8.1 Discussion

The simplest choice is the same polynomial degree for velocity and pressure, for instance $\mathbb{P}_1$ for both. It is easy to implement and gives a small system. It is also wrong, and the way it fails is instructive.

There are two separate obstructions. The first is a counting obstruction. The inf-sup condition says every discrete pressure is detected by some discrete velocity, which forces the adjoint of the map induced by b to be injective, which in turn forces $\dim Q_h \leq \dim \mathbf{V}_h$. If there are more pressure unknowns than velocity unknowns the condition fails for purely dimensional reasons. This is necessary and easy to check, but not sufficient.

The second obstruction is what actually kills $\mathbb{P}_1/\mathbb{P}_1$. On a regular mesh there exists a specific nonzero pressure function, alternating in sign between neighbouring vertices like the squares of a checkerboard, which is very nearly orthogonal to the divergence of every discrete velocity. It is not exactly orthogonal, but the pairing is small relative to the norms and becomes smaller under refinement, so $S_h \rightarrow 0$ and the pair is not uniformly stable.

The practical symptom is easy to recognise. The computed velocity may look entirely reasonable while the computed pressure is polluted by a large oscillation with the checkerboard pattern. This is not a bug in the code. It is the spurious mode, present in the discrete problem itself, and refining the mesh does not remove it.

8.2 Mathematics

Proposition 8.1 (Necessary dimension condition). Define $B_h : \mathbf{V}_h \rightarrow Q'_h$ by $(B_h \mathbf{v}_h)(q_h) = b(\mathbf{v}_h, q_h)$. If (29) holds then $B_h^* : Q_h \rightarrow \mathbf{V}'_h$ is injective, and consequently $\dim Q_h \leq \dim \mathbf{V}_h$.

Proof. If $\beta_* > 0$ and $q_h \neq 0$ then $\sup_{\mathbf{v}_h} b(\mathbf{v}_h, q_h) > 0$, so $B_h^* q_h \neq 0$. Hence $\ker B_h^* = \{0\}$ and $\dim Q_h = \dim \text{ran } B_h^* \leq \dim \mathbf{V}'_h = \dim \mathbf{V}_h$. $\square$

Proposition 8.2 (Dimension count for $\mathbb{P}_1/\mathbb{P}_1$). Let N_{int} and N_{tot} be the numbers of interior and of all vertices of $\mathcal{T}_h$. For $\mathbf{V}_h = [\mathcal{L}_1^0]^2$ and $Q_h = \mathcal{L}_1 \cap L_0^2(\Omega)$,

$$\dim \mathbf{V}_h = 2N_{\text{int}}, \quad \dim Q_h = N_{\text{tot}} - 1.$$

Proposition 8.3 (Checkerboard mode). Let $\Omega = (0, 1)^2$ be meshed by an $n \times n$ uniform grid of squares, each divided into two triangles, with vertices $(x_i, y_j) = (i/n, j/n)$. Let $\mathbf{V}_h = [\mathcal{L}_1^0]^2$, $Q_h = \mathcal{L}_1 \cap L_0^2(\Omega)$, and let $q_h^* \in \mathcal{L}_1$ be determined by

$$q_h^*(x_i, y_j) = (-1)^{i+j}, \quad 0 \leq i, j \leq n,$$

adjusted by its mean so that $q_h^* \in Q_h$. Then

$$\sup_{\mathbf{v}_h \in \mathbf{V}_h \setminus \{0\}} \frac{b(\mathbf{v}_h, q_h^*)}{\|\mathbf{v}_h\|_{H^1} \|q_h^*\|_{L^2}} \longrightarrow 0 \quad (h \rightarrow 0),$$

so (29) fails.

Corollary 8.4. *By Theorem 10.1 the pressure error bound is proportional to β_*^{-1} . With $S_h \rightarrow 0$ the bound degenerates and p_h admits a component $c q_h^*$ with c not controlled by the data, while $\mathbf{u}_h$ may remain accurate.*

9 Stability of Taylor–Hood

9.1 Discussion

The repair is to make the velocity space richer than the pressure space. Taking $\mathbb{P}_2$ for velocity and $\mathbb{P}_1$ for pressure gives the velocity enough freedom to detect every discrete pressure. The resulting pair, called Taylor–Hood, is uniformly stable, is the standard choice, and is what we implement.

Stability is not proved by direct estimation. The standard route is Fortin’s criterion, which converts the problem into the construction of a single operator. The idea is this. The continuous inf-sup condition already supplies, for each discrete pressure, a continuous velocity detecting it. What is missing is a discrete velocity. So it suffices to build a map Π_h sending continuous velocities to discrete ones in such a way that the pairing against discrete pressures is unchanged and the norm grows by at most a fixed factor. If such a map exists, the continuous condition can be pushed down to the discrete spaces with the constant degraded only by that factor.

The two requirements on Π_h are exactly these: preservation of the pairing against Q_h , and stability in the $H^1(\Omega)$ norm with an h -independent constant. The construction of Π_h for Taylor–Hood is technical, and it is where the requirement that the mesh contain at least three triangles enters, that requirement excluding degenerate cases where the pressure space is too large relative to the velocity space. I quote the construction and prove only the criterion, which is short.

9.2 Mathematics

Lemma 9.1 (Fortin’s criterion). *Suppose (23) holds with constant β , and suppose there exist a linear operator $\Pi_h : \mathbf{V} \rightarrow \mathbf{V}_h$ and a constant $C_\Pi > 0$ independent of h with*

$$b(\Pi_h \mathbf{v} - \mathbf{v}, q_h) = 0 \quad \forall \mathbf{v} \in \mathbf{V}, \forall q_h \in Q_h, \quad (30)$$

$$\|\Pi_h \mathbf{v}\|_{H^1} \leq C_\Pi \|\mathbf{v}\|_{H^1} \quad \forall \mathbf{v} \in \mathbf{V}. \quad (31)$$

Then (29) holds with $\beta_* \geq \beta/C_\Pi$.

Proof. Fix $q_h \in Q_h \setminus \{0\}$. By (23) choose $\mathbf{v} \in \mathbf{V} \setminus \{0\}$ with $b(\mathbf{v}, q_h) \geq \beta \|\mathbf{v}\|_{H^1} \|q_h\|_{L^2}$. By (30), $b(\Pi_h \mathbf{v}, q_h) = b(\mathbf{v}, q_h)$, and by (31), $\|\Pi_h \mathbf{v}\|_{H^1} \leq C_\Pi \|\mathbf{v}\|_{H^1}$. Hence

$$\frac{b(\Pi_h \mathbf{v}, q_h)}{\|\Pi_h \mathbf{v}\|_{H^1} \|q_h\|_{L^2}} = \frac{b(\mathbf{v}, q_h)}{\|\Pi_h \mathbf{v}\|_{H^1} \|q_h\|_{L^2}} \geq \frac{\beta \|\mathbf{v}\|_{H^1} \|q_h\|_{L^2}}{C_\Pi \|\mathbf{v}\|_{H^1} \|q_h\|_{L^2}} = \frac{\beta}{C_\Pi}.$$

Since $\Pi_h \mathbf{v} \in \mathbf{V}_h$, the supremum over $\mathbf{V}_h$ is at least β/C_Π , and q_h was arbitrary. $\square$

Theorem 9.2 (Taylor–Hood is inf-sup stable). *Let $\{\mathcal{T}_h\}$ be shape regular in the sense of (25), each $\mathcal{T}_h$ containing at least three triangles, and let $\mathbf{V}_h, Q_h$ be as in (26). Then (29) holds with $\beta_* > 0$ depending only on Ω and σ .*

The operator Π_h satisfying (30) and (31) is constructed in Boffi, Brezzi and Fortin; the result is quoted.

Corollary 9.3. *β_* is independent of ν, ν_r, c_a, c_d , since b does not involve these parameters.*

10 Error estimates

10.1 Discussion

With stability established the remaining question is accuracy. The general principle, going back to Céa's lemma for coercive problems, is that a conforming discretisation is quasi-optimal: the error of the computed solution is bounded by the best error achievable in the discrete space, times a constant depending only on the stability constants. The method cannot do much better than the space allows, and stability guarantees it does not do much worse.

The saddle point version is estimate (32) below. It contains two best approximation terms, one for the velocity and microrotation pair and one for the pressure, and two amplification factors, M/α from the coercive block and $\|b\|/\beta_*$ from the constraint block. The second factor is why inf-sup stability matters quantitatively and not merely qualitatively: as β_* degrades the bound worsens even though a solution still exists.

The best approximation terms are then estimated by interpolation theory, which says how well a smooth function is approximated by piecewise polynomials of a given degree on a mesh of given size. This is where shape regularity is used, and where the degrees $k = 2$ and $k = 1$ produce the rates.

One point deserves emphasis. The rates come from the interpolation estimates and depend only on the polynomial degrees and the smoothness of the exact solution. The constants come from the stability estimates and depend on the physical parameters. This separation is what makes the parameter study of Section 11 meaningful: varying ν_r moves the constants but not the rates, and the two are independently observable.

10.2 Mathematics

Theorem 10.1 (Discrete Brezzi and quasi-optimality). *Assume a bounded with constant M and coercive on Z_h with constant α , b bounded with constant $\|b\| = \sqrt{2}$, and (29) with constant β_* . Then (27)–(28) has a unique solution and*

$$\begin{aligned} & \|(\mathbf{u}, w) - (\mathbf{u}_h, w_h)\| + \|p - p_h\|_{L^2} \\ & \leq C \left(1 + \frac{M}{\alpha}\right) \left(1 + \frac{\|b\|}{\beta_*}\right) \left[\inf_{(\mathbf{v}_h, z_h) \in \mathbf{V}_h \times W_h} \|(\mathbf{u}, w) - (\mathbf{v}_h, z_h)\| + \inf_{q_h \in Q_h} \|p - q_h\|_{L^2} \right], \end{aligned} \quad (32)$$

with C absolute.

Lemma 10.2 (Interpolation). *Let $\{\mathcal{T}_h\}$ be shape regular and I_h^k the Lagrange interpolant into $\mathcal{L}_k$. For $\phi \in H^{k+1}(\Omega)$ and $m \in \{0, 1\}$,*

$$\|\phi - I_h^k \phi\|_{H^m(\Omega)} \leq C h^{k+1-m} |\phi|_{H^{k+1}(\Omega)}, \quad C = C(\sigma, k, m). \quad (33)$$

Theorem 10.3 (Convergence rates). *Let $\mathbf{u} \in [H^3(\Omega)]^2 \cap \mathbf{V}$, $w \in H^3(\Omega) \cap W$, $p \in H^2(\Omega) \cap Q$, with $\mathbf{V}_h, W_h, Q_h$ as in (26) on a shape regular family. Then*

$$\|\nabla(\mathbf{u} - \mathbf{u}_h)\|_{L^2} + \|\nabla(w - w_h)\|_{L^2} + \|p - p_h\|_{L^2} \leq C_* h^2, \quad (34)$$

$$\|\mathbf{u} - \mathbf{u}_h\|_{L^2} + \|w - w_h\|_{L^2} \leq C_* h^3, \quad (35)$$

where

$$C_* = C \left(1 + \frac{M}{\alpha}\right) \left(1 + \frac{\sqrt{2}}{\beta_*}\right) (|\mathbf{u}|_{H^3} + |w|_{H^3} + |p|_{H^2}). \quad (36)$$

Proof. For (34), bound the two infima in (32) by the particular choices $\mathbf{v}_h = I_h^2 \mathbf{u}$, $z_h = I_h^2 w$, $q_h = I_h^1 p$, and apply (33) with $(k, m) = (2, 1)$ for the first and $(k, m) = (1, 0)$ for the second. Both give $O(h^2)$. Substituting into (32) gives the claim.

Estimate (35) follows by the Aubin–Nitsche duality argument, which requires H^2 regularity of the dual problem; this holds for convex Ω . $\square$

11 Parameter dependence

11.1 Discussion

The vortex viscosity ν_r enters M but not α , by Lesson 2, and by Corollary 9.3 it does not enter β_* either. So the whole parameter dependence of the error is carried by the single factor M/α , which grows linearly in ν_r .

This gives two predictions that are independently testable and that separate rates from constants. In a mesh refinement study at fixed ν_r the observed exponents should be 2 and 3 regardless of the value of ν_r . In a parameter sweep at fixed mesh the errors should grow linearly once ν_r dominates the other coefficients.

The quantity M/α is also the natural condition number of the problem, so the same growth predicts that iterative solvers require more iterations at large ν_r , and that preconditioning becomes necessary rather than optional.

11.2 Mathematics

Proposition 11.1.

$$\frac{M}{\alpha} = \frac{(\nu + (c_a + c_d) + 9\nu_r)(C_\Omega^2 + 1)}{\min(\nu, c_a + c_d)} = O(\nu_r) \quad (\nu_r \rightarrow \infty),$$

and by Corollary 9.3 together with (36),

$$C_* = O(\nu_r) \quad (\nu_r \rightarrow \infty),$$

while the exponents in (34) and (35) are unchanged.

Corollary 11.2 (Testable predictions). *(P1) At fixed ν_r , the observed rates are 2 in H^1 and 3 in $L^2(\Omega)$, independently of ν_r .*

(P2) At fixed h and $\nu_r \gg \max(\nu, c_a + c_d)$, the errors grow as $O(\nu_r)$.

12 The linear system

12.1 Discussion

Choosing bases for the three discrete spaces turns (27) and (28) into a matrix equation whose structure mirrors that of the continuous problem: a symmetric block for the coercive part, an off diagonal block for the constraint, and a zero block in the pressure to pressure position because the pressure never pairs with itself. That zero block is the algebraic form of (22).

The zero block also makes the matrix indefinite, meaning it has eigenvalues of both signs. The immediate practical consequence is that the conjugate gradient method, which requires positive definiteness, cannot be used. Direct factorisation is appropriate at moderate size; beyond that one uses MINRES or GMRES with a block preconditioner, whose quality degrades as M/α grows.

Nonsingularity of the matrix is the algebraic statement of the two Brezzi conditions: the velocity block must be coercive on the kernel of the constraint, and the constraint block must have injective transpose. The second is precisely where the zero mean condition on the pressure space acts, since without it the constant vector lies in the kernel of the transpose and the matrix is singular.

12.2 Mathematics

Let $\{\varphi_i\}_{i=1}^{n_u}$, $\{\psi_j\}_{j=1}^{n_w}$, $\{\chi_k\}_{k=1}^{n_p}$ be bases of $\mathbf{V}_h$, W_h , Q_h , and write $\mathbf{u}_h = \sum_i U_i \varphi_i$, $w_h = \sum_j W_j \psi_j$, $p_h = \sum_k P_k \chi_k$. Then

$$\begin{pmatrix} A_{uu} & A_{uw} & B^\top \\ A_{wu} & A_{ww} & 0 \\ B & 0 & 0 \end{pmatrix} \begin{pmatrix} U \\ W \\ P \end{pmatrix} = \begin{pmatrix} F_u \\ F_w \\ 0 \end{pmatrix}, \quad (37)$$

$$\begin{aligned} (A_{uu})_{ij} &= (\nu + \nu_r) (\nabla \varphi_j, \nabla \varphi_i), & (A_{uw})_{ij} &= -2\nu_r (\nabla^\perp \psi_j, \varphi_i), \\ (A_{wu})_{ij} &= -2\nu_r (\text{curl } \varphi_j, \psi_i), & (A_{ww})_{ij} &= (c_a + c_d) (\nabla \psi_j, \nabla \psi_i) + 4\nu_r (\psi_j, \psi_i), \\ B_{ki} &= -(\chi_k, \text{div } \varphi_i). \end{aligned}$$

Proposition 12.1 (Symmetry). $A_{uw} = A_{wu}^\top$, so the matrix in (37) is symmetric.

Proof. By Lemma 4.4, $(\nabla^\perp \psi_j, \varphi_i) = (\psi_j, \text{curl } \varphi_i) = (\text{curl } \varphi_i, \psi_j)$. Multiplying by $-2\nu_r$ gives $(A_{uw})_{ij} = (A_{wu})_{ji}$. $\square$

Proposition 12.2 (Indefiniteness). The matrix in (37) has eigenvalues of both signs.

Proof. Let $\mathcal{M}$ denote the matrix and let e be a unit vector supported on the pressure block. Then $e^\top \mathcal{M} e = 0$ by the zero block, while $v^\top \mathcal{M} v > 0$ for $v \neq 0$ supported on the velocity block, by Proposition 7.3. A symmetric matrix possessing a nonzero isotropic vector together with a positive direction is indefinite. $\square$

Proposition 12.3 (Nonsingularity). (37) is nonsingular if and only if $\ker B^\top = \{0\}$ and a is coercive on $\ker B$. The first is the algebraic form of (29), the second is Proposition 7.3. The restriction $Q_h \subset L_0^2(\Omega)$ removes the constant vector from $\ker B^\top$.

13 Verification

13.1 Discussion

The method of manufactured solutions tests the implementation against the theory. One chooses fields satisfying the boundary conditions and the constraint, substitutes them into the strong form to obtain the data $\mathbf{f}$ and g that they solve, runs the code with that data, and compares the computed solution against the known exact one. If the observed convergence rates match Theorem 10.3 the discretisation is correct; if they do not there is a bug.

The test is sharp because the rates are integers predicted in advance. A rate of 2.4 where the theory says 3 is not a partial success, it is a defect. The most common causes, in order of frequency, are quadrature of insufficient degree, computing the error against an interpolant of too low degree, a sign error in a coupling term, and boundary conditions applied to the wrong component of the mixed space.

The second of these deserves care. If the error is computed by interpolating the exact solution into the same space $\mathcal{L}_k$ used for the solution, one measures the interpolation error rather than the discretisation error, and the observed rates plateau for reasons unrelated to the scheme.

13.2 Mathematics

On $\Omega = (0, 1)^2$ define

$$\psi(x, y) = \sin^2(\pi x) \sin^2(\pi y), \quad (38)$$

$$\mathbf{u} = \nabla^\perp \psi = (\partial_2 \psi, -\partial_1 \psi), \quad (39)$$

$$p = \cos(\pi x) \cos(\pi y), \quad (40)$$

$$w = \sin(\pi x) \sin(\pi y). \quad (41)$$

Proposition 13.1 (Admissibility). $\mathbf{u} \in \mathbf{V}_{\text{div}}, p \in Q, w \in W$.

Proof. $\text{div } \nabla^\perp \psi = \partial_1 \partial_2 \psi - \partial_2 \partial_1 \psi = 0$. Since ψ and $\nabla \psi$ vanish on $\partial\Omega$, so does $\mathbf{u}$; likewise $w = 0$ on $\partial\Omega$. Finally

$$\int_0^1 \int_0^1 \cos(\pi x) \cos(\pi y) \, dx \, dy = \left(\int_0^1 \cos(\pi s) \, ds \right)^2 = \left(\left[\frac{1}{\pi} \sin(\pi s) \right]_0^1 \right)^2 = 0. \quad \square$$

Define the data by substitution into (11) and (13):

$$\mathbf{f} := -(\nu + \nu_r) \Delta \mathbf{u} + \nabla p - 2\nu_r \nabla^\perp w, \quad (42)$$

$$g := -(c_a + c_d) \Delta w + 4\nu_r w - 2\nu_r \text{curl } \mathbf{u}. \quad (43)$$

On meshes $h_m = 2^{-m}$, $m = 2, \dots, 6$, compute

$$e_m^X = \|X - X_{h_m}\|, \quad r_m^X = \frac{\log(e_{m-1}^X / e_m^X)}{\log 2}. \quad (44)$$

Quantity	Norm	Expected r_m
$\mathbf{u} - \mathbf{u}_h$	$L^2(\Omega)$	3
$\mathbf{u} - \mathbf{u}_h$	H^1	2
$p - p_h$	$L^2(\Omega)$	2
$w - w_h$	$L^2(\Omega)$	3
$w - w_h$	H^1	2

Errors are computed against $I_h^{k+2} X$ with quadrature of degree at least $2k + 2$.

Parameter test: fix h and record e^X and r^X for $\nu_r/\nu \in \{10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3\}$, against predictions (P1) and (P2) of Corollary 11.2.

14 Summary

Property	Transfers to $\mathbf{V}_h, W_h, Q_h$?	Reference
Boundedness of a and b	yes	Prop. 7.1
Coercivity	yes	Prop. 7.3
$Z_h \subset Z$	no	Prop. 7.2
Inf-sup, general pair	no	Prop. 7.4
Inf-sup, $\mathbb{P}_1/\mathbb{P}_1$	no	Prop. 8.3
Inf-sup, $\mathbb{P}_2/\mathbb{P}_1$	yes	Thm. 9.2